Minor Project 3

Vigneswaran Madappan Chinnasami

1. Project Overview

Fitness Manager is a gym management web application developed with ASP.NET Core Razor Pages and C#. It uses a MySQL database and a separate class library for data access. The application allows users to manage clients, trainers, and exercises, search client records, switch between light and dark themes, and view an administration dashboard.

2. Technologies Used

• ASP.NET Core Razor Pages (.NET 10) — web interface

• C# — application and data-access logic

• MySQL — relational database

• MySqlConnector — connection between C# and MySQL

• Bootstrap — responsive styling and theme support

• Visual Studio Code on macOS — development environment

3. Database Design

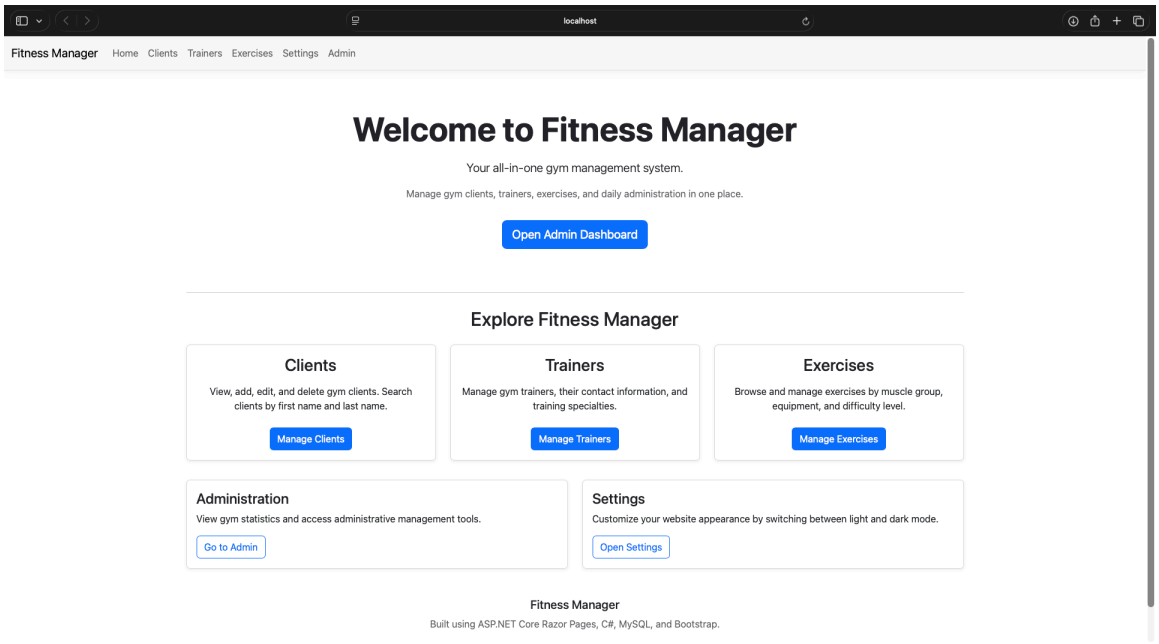
Database name: fitness_manager. The database contains five tables and 31 records at final verification.

Table	Verified records
Clients	6
Trainers	6
TrainingPlans	5
Exercises	13
Settings	1
TOTAL	31

The database creation and insertion scripts are included as Scripts/DDL.sql and Scripts/DML.sql. The application uses a separate MinorProject3.Data class library for database access.

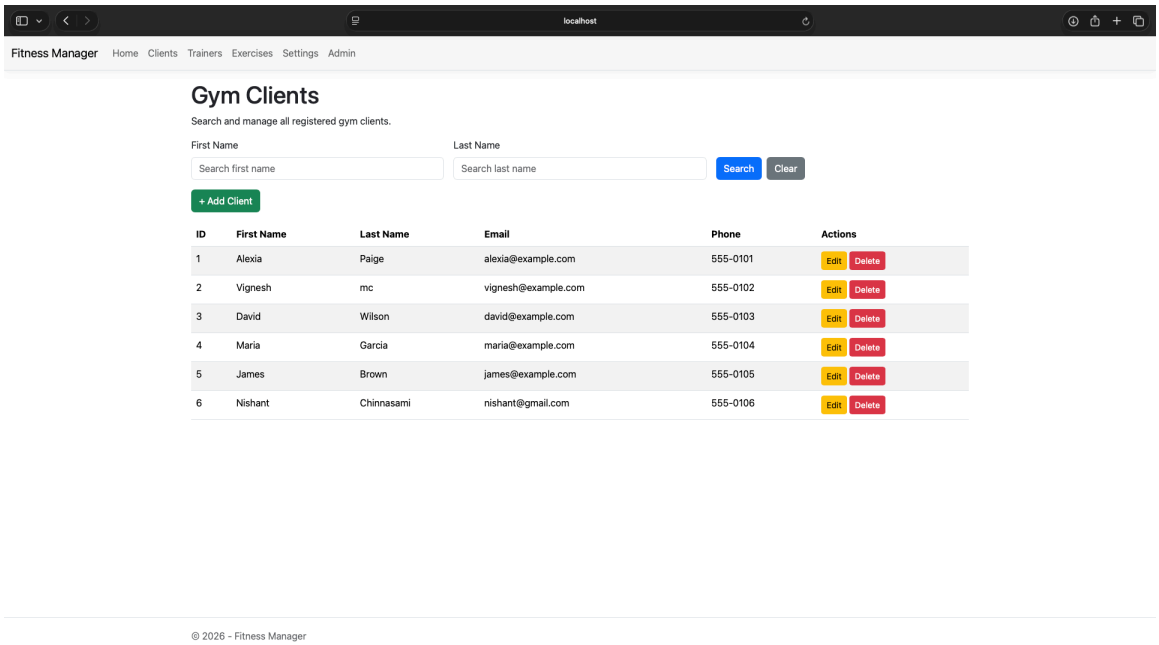
4. Homepage and Navigation

The homepage provides entry points to Clients, Trainers, Exercises, Administration, and Settings. The navigation bar provides access to the major pages.



5. Clients: Read and Search

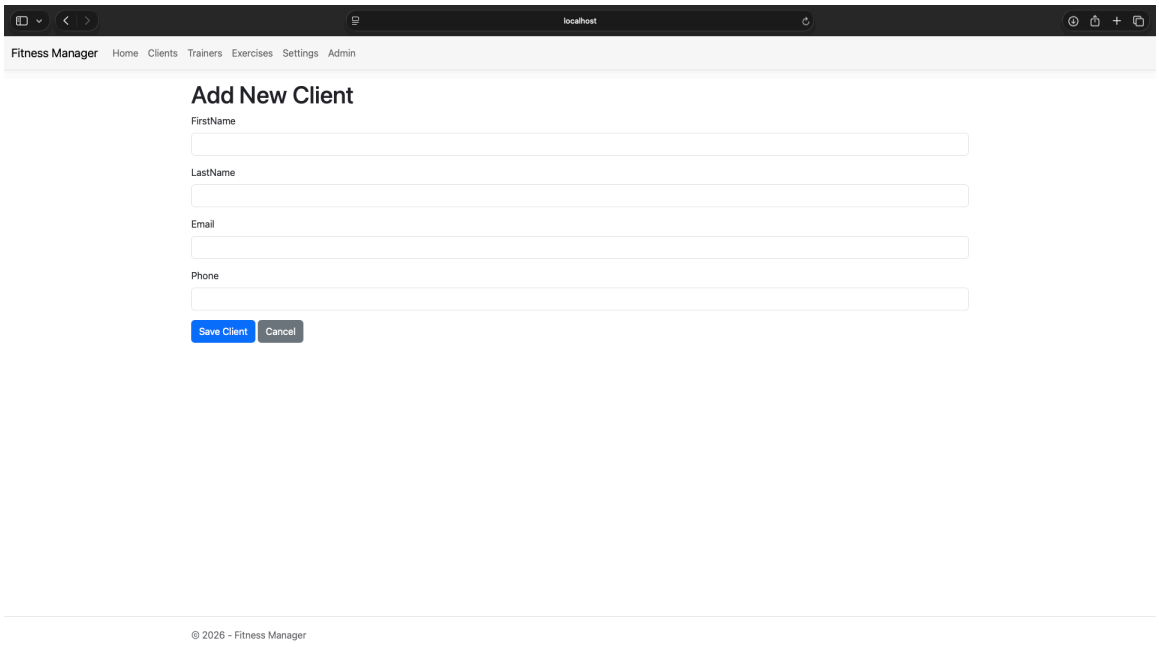
The Clients page lists existing clients. Two separate text inputs search FirstName and LastName using parameterized SQL LIKE queries. The Clear action restores the full list.



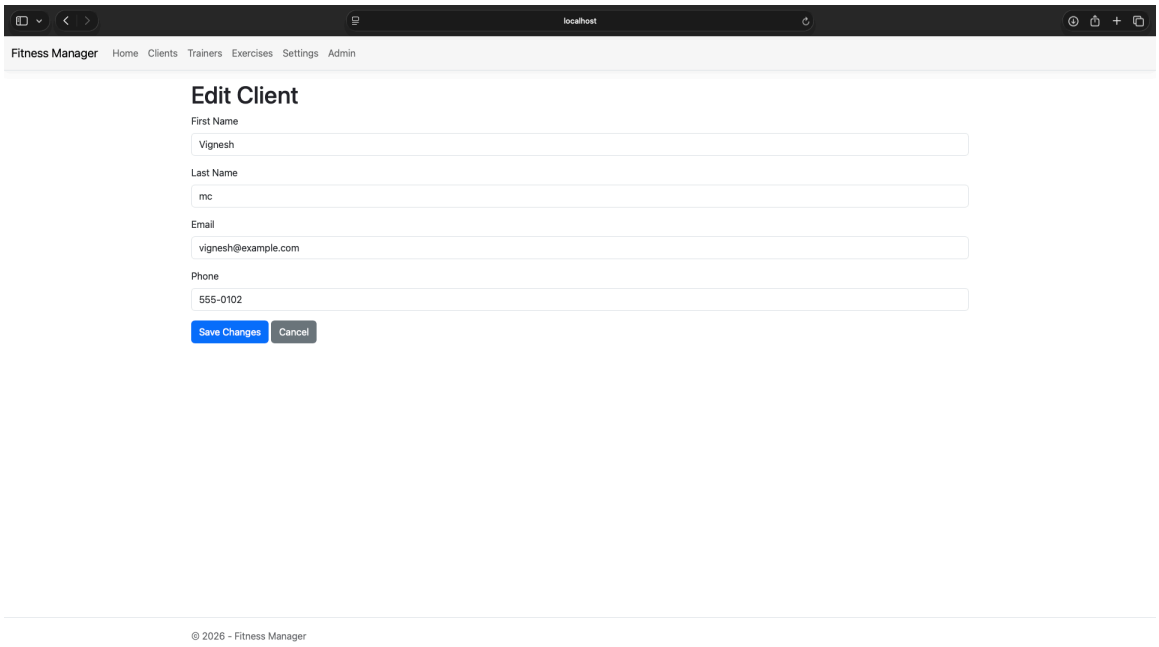
6. Client CRUD Operations

The application supports Create, Read, Update, and Delete operations on client records. The screenshots below document the operations. Deletion of records referenced by training plans is blocked by the database foreign-key relationship.

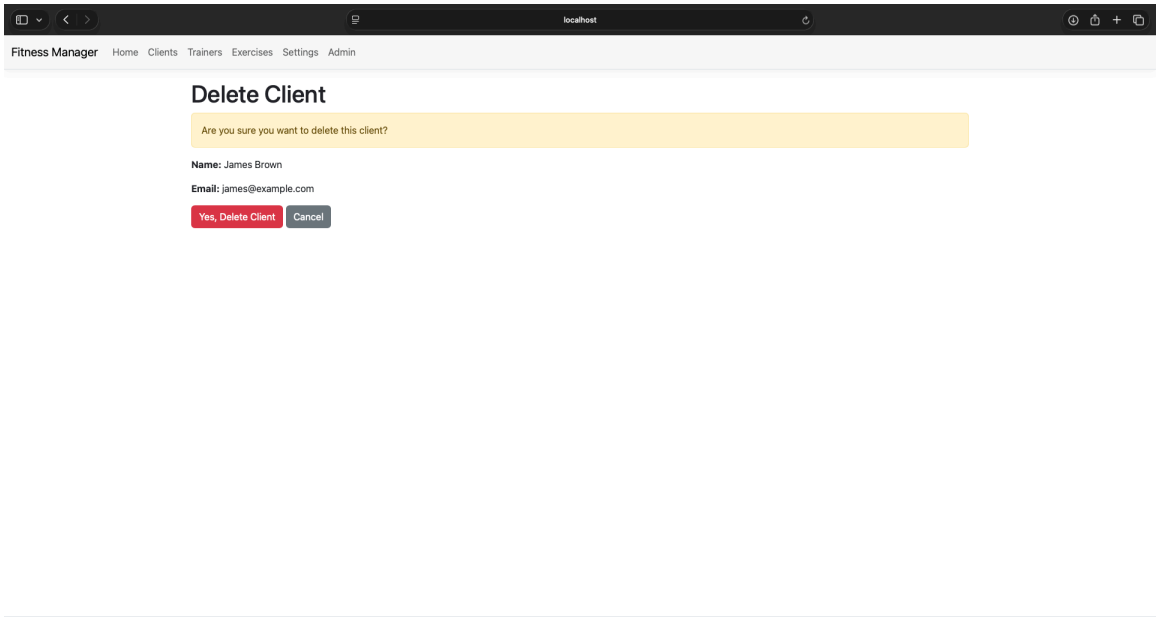
Create a Client



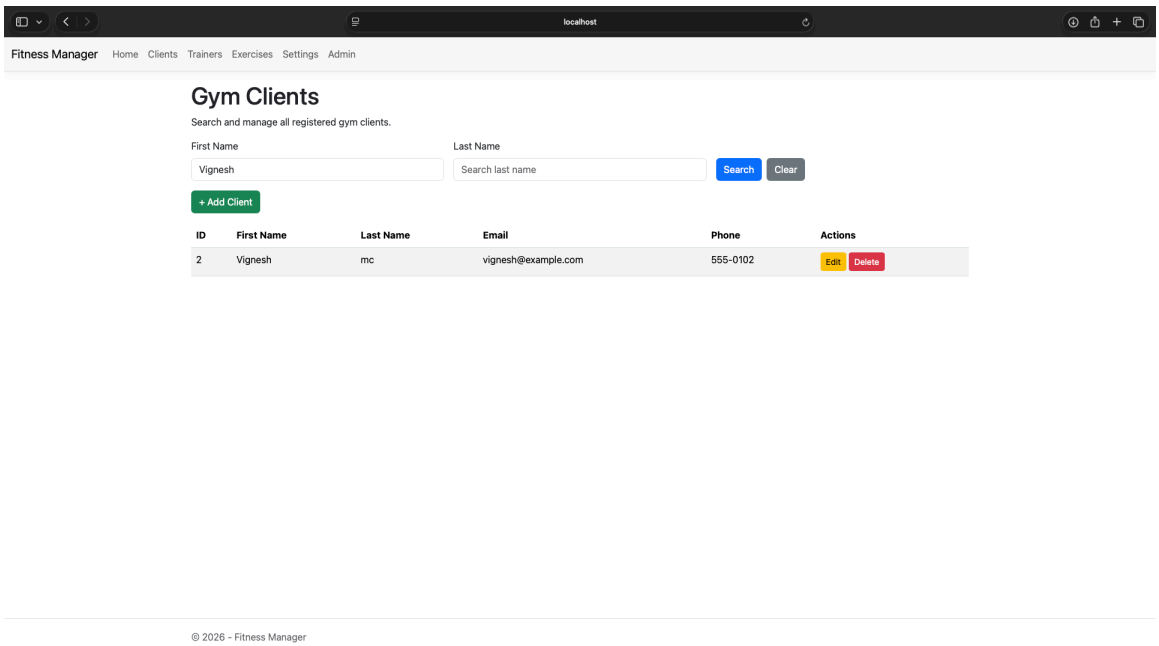
Update a Client



Delete a Client

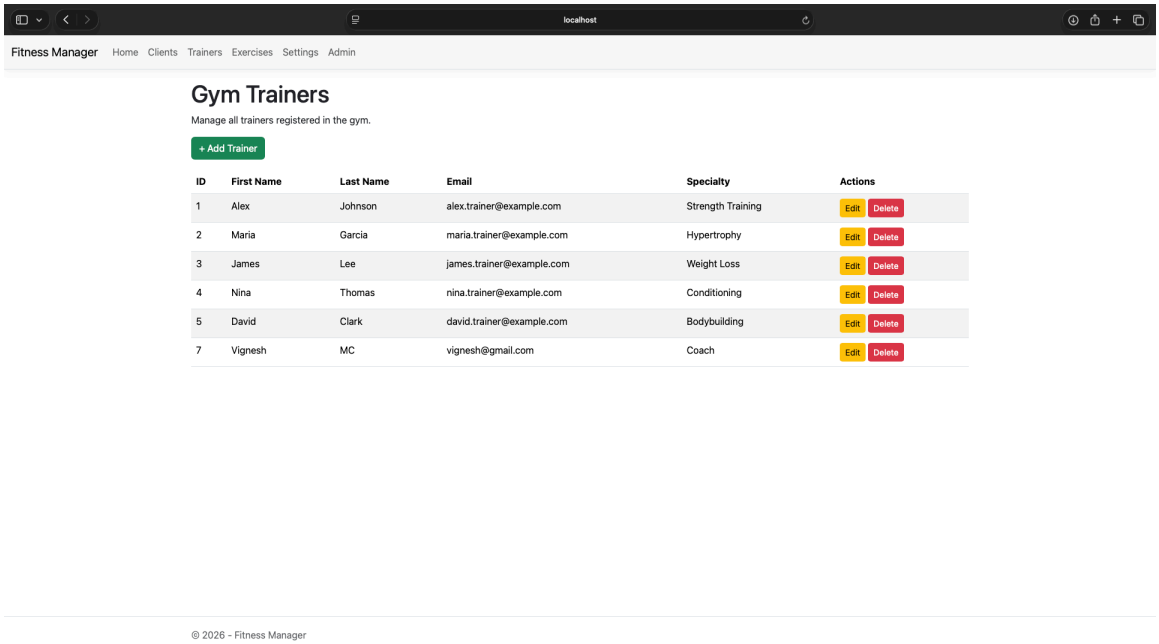


Search by Two Text Fields



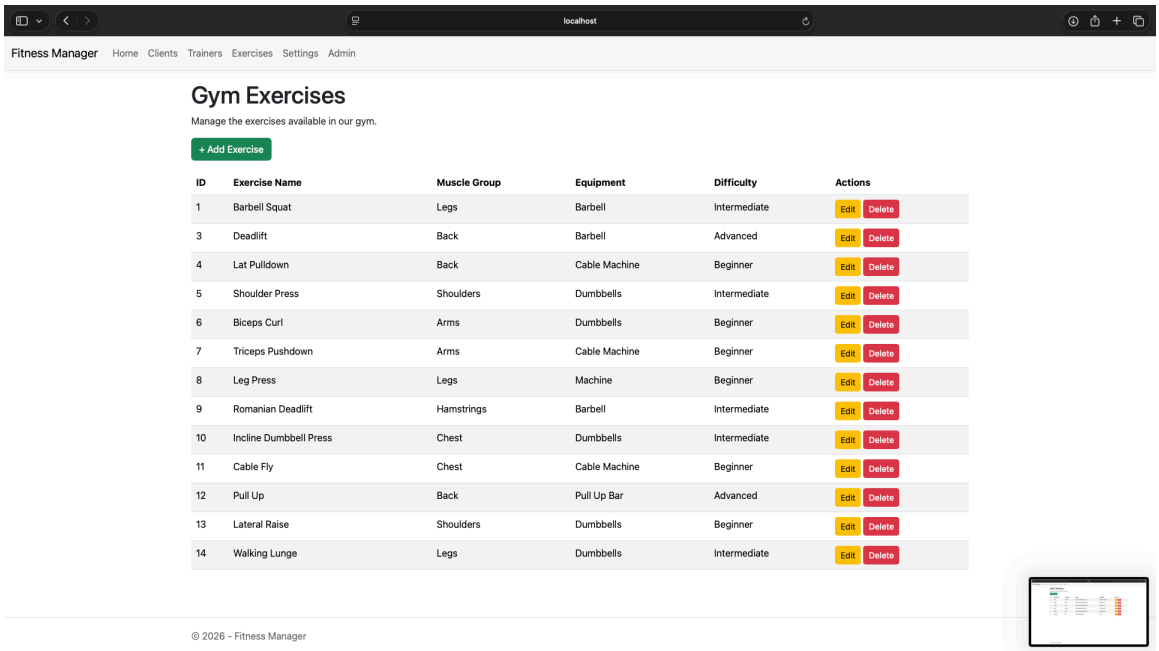
7. Trainers CRUD

The Trainers page supports listing, adding, editing, and deleting trainer records, including their contact details and specialties. The operations use TrainerRepository in the separate data layer.



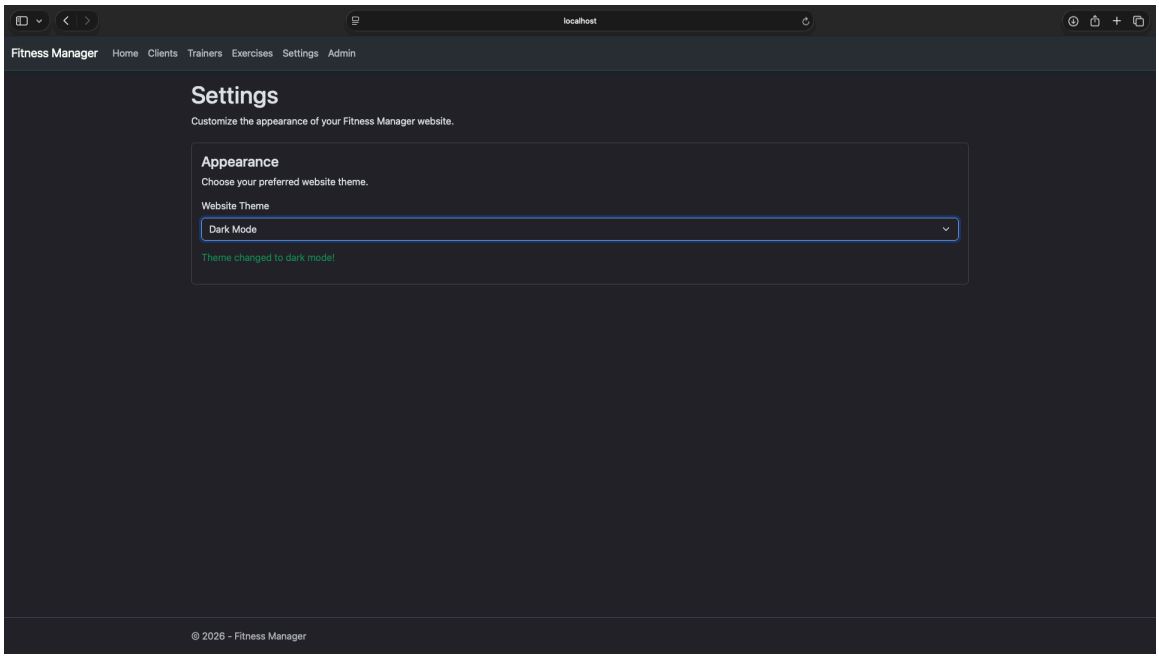
8. Exercises CRUD

The Exercises page supports listing, adding, editing, and deleting exercises. Each exercise stores its name, muscle group, equipment, and difficulty level. The operations use ExerciseRepository in the separate data layer.



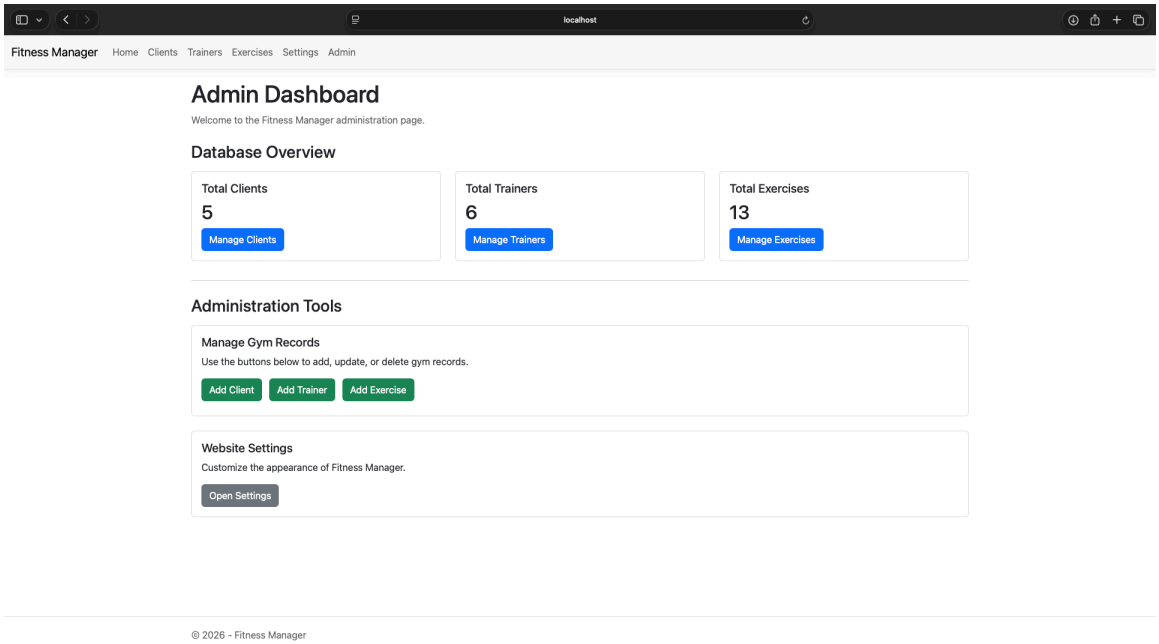
9. Settings: Light and Dark Mode

The Settings page allows the user to switch between light and dark mode. The theme updates immediately across the website and the browser saves the preference using localStorage.



10. Administration Dashboard

The Admin Dashboard displays live counts of clients, trainers, and exercises, with shortcuts to manage records and open Settings. This page is an administrative interface; login-based authorization was not implemented.



11. Testing and Verification

• The solution builds successfully using dotnet build.

• Client, Trainer, and Exercise Create, Read, Update, and Delete operations were tested.

• Client search using FirstName and LastName was tested.

• Light/dark theme switching and persistence were tested.

• The Admin Dashboard and navigation links were tested.

• MySQL verification confirmed 5 tables and 31 total records.

12. Project Structure

• MinorProject3.Web/ — Razor Pages, layouts, UI and configuration

• MinorProject3.Data/ — models, repositories, and database connection factory

• Scripts/DDL.sql — database table definitions

• Scripts/DML.sql — exported sample records

13. Conclusion

Fitness Manager demonstrates a working ASP.NET Core Razor Pages application backed by MySQL. It includes a separate data layer, CRUD operations on three tables, two-field search, a working theme preference, and an administration dashboard. The project meets the verified database size requirement of five tables and at least thirty records.